

UG" END SEMESTER EXAMINATION – JANUARY, 2025

(REGULAR-ODD SEMESTER)

1st YEAR (1st SEMESTER) - MECHANICAL ENGINEERING

ENGINEERING MECHANICS

MENUGES01

(Full Marks – 80)

Time – 3 Hours

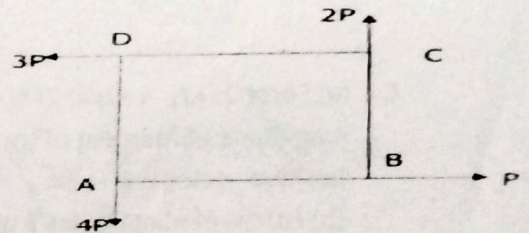
(Assume any missing data if any)

Group- A

(Answer any five questions) [5x (3+5) =40]

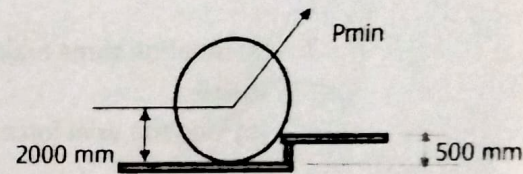
1. (a) What is transmissibility of force? Explain with example. Is it applicable to internal forces also?

(b) Four forces equal to P , $2P$, $3P$ & $4P$ are respectively acting along the four sides of a square ABCD taken in order (As Shown). Find the magnitude, direction and position of the resultant force if each side of square is 100mm.



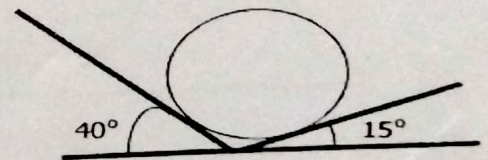
2. (a) State Lami's theorem.

(b) A roller of radius 2000mm and weight 100N is to be pulled over a curb of height 500mm by a horizontal force 'P' applied to end of a string wound round the circumference of the roller as shown. What will be the minimum magnitude of force 'P' required to start the roller over the curb?



3. (a) What is force system & mention types of 2D force system with sketches.

(b) A smooth circular cylinder of radius 1.5m is lying in a triangular groove. One side of which makes 15° angle and the other 40° angle with horizon. Find the reactions at the surface of contact, if there is no friction and the cylinder weight is 100N.



4. (a) What are moment of Force and moment of couple ? Explain force couple system with an example.

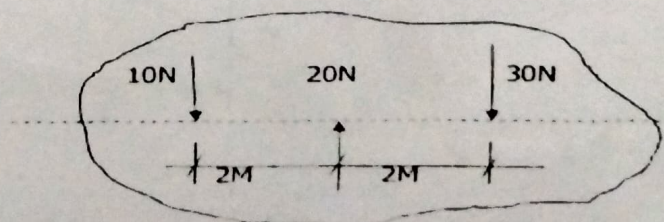
(b) For given many forces on the body find

(i) Resultant Force R

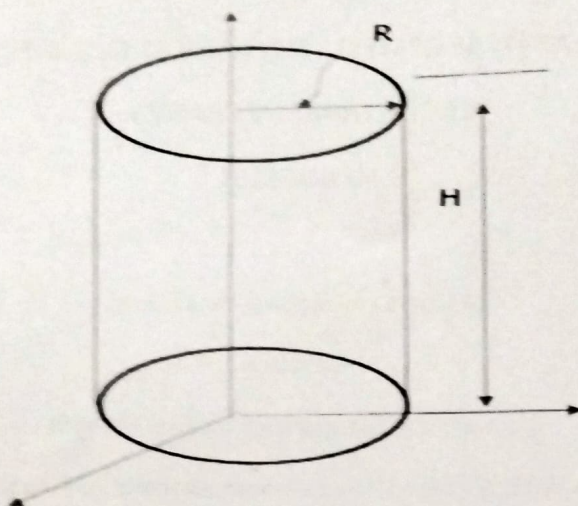
(ii) Net Resultant moment at A, B, C

(iii) Net- equivalent force couple system at A, B, C

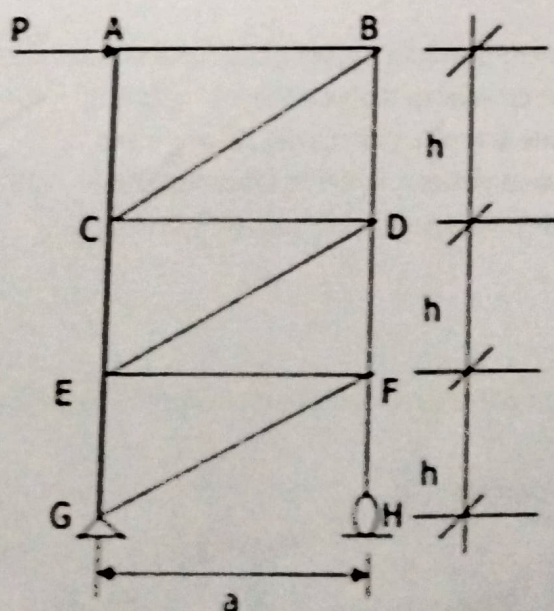
(iv) the position of resultant R with respect to A, B, C



5. (a) State parallel axis theorem in moment of inertia.
 (b) Determine the mass moment of Inertia of the Cylinder of radius, R & height, H .



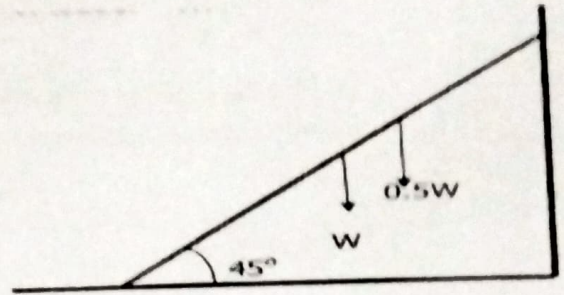
6. (a) Force $2\mathbf{i} + 7\mathbf{j}$, $-\mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $2\mathbf{i} - 5\mathbf{j} + 6\mathbf{k}$ act at a point $P(4\mathbf{i} - 3\mathbf{j} - 2\mathbf{k})$. Find the vector moment and the magnitude of moment of the resultant of these forces acting at 'P' about the point 'O' whose position vector is $6\mathbf{i} + \mathbf{j} - 3\mathbf{k}$.
 (b) Forces of magnitudes 5 units and 3 units acting in the direction $6\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ and $3\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$ respectively act on a particle which is displaced from the point $(2, 2, -1)$ to $(4, 3, 1)$. Find the work done by the forces.
7. (a) Mention three major differences between truss and frame. What is imperfect and perfect truss?
 (b) Find the axial forces in members CE, ED & DF by section method.



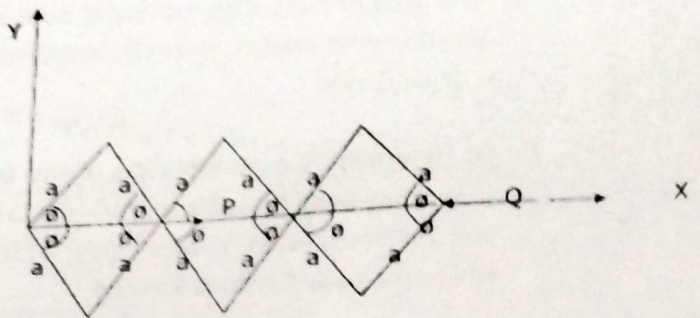
Group – B

(Answer any four questions only) [4x (10) =40]

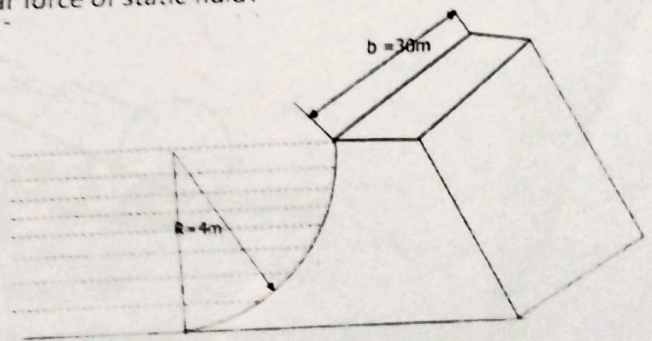
1. (a) Define friction. State three dry coulomb's Laws.
 (b) A uniform Ladder of 4m length rests against a vertical wall with which it makes an angle of 45° . The co-efficient of between ladder and the wall is 0.4 and that between floor and the ladder is 0.5. Now, if man whose weight is one half of that of the ladder ascends it then how high will it be when the ladder is just starting to slip? Weight of ladder = W .



2. (a) State principle of virtual work.
 (b) Find the relationship between P and Q for equilibrium of the system as shown in figure. All the members are assumed to be weightless. Apply virtual work principle to solve this problem. Each side of the structure having a length of ' a ' and makes an angle θ with ' X ' axis as shown in figure.

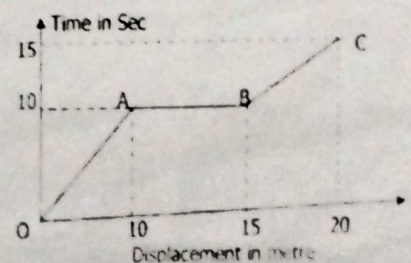


3. (a) What is static fluid? What will be the shear force of static fluid?
 (b) Determine completely the resultant force ' R ' exerted on the cylindrical dam surface by the water. The density of water is 1000 kg/m^3 and the dam has a length of ' $b=30\text{m}$ ' normal to the paper.

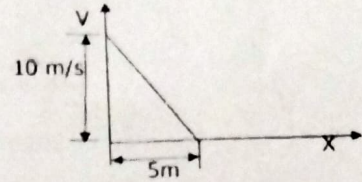


4. (a) A ball is thrown vertically upward from ground. Two men standing on different height of a building watch the ball passing by the at 20m/sec and 10 m/sec respectively. How high does the ball rise beyond second man and first man?
 (b) A particle moving in a plane as $(x=2+3t^2, y= 3+t^3)$, then find its position, velocity and acceleration at $t=3 \text{ sec}$.

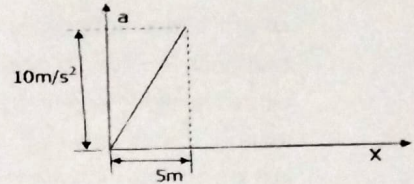
5. (a) Time displacement graph is given below. Find the velocity in OA, AB and BC region.



(b) Find acceleration at $x=2\text{m}$ from given X (Displacement)- V Velocity graph.



(c) Find the speed of the particle at $X=5\text{m}$ if at $X=0$, speed is 2m/s . Given a (Acceleration) - X (Displacement) graph.

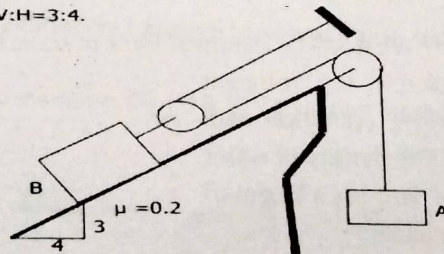


6. (a) Define impact and collision.

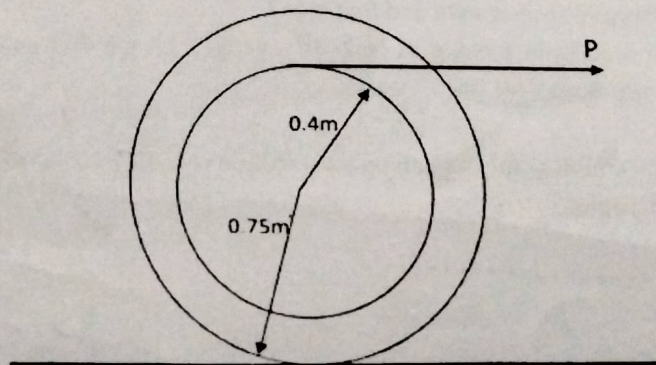
(b) A body of mass 30kg moving at 3m/s strike to another body of mass 15kg in just opposite direction with 7m/sec . If co-efficient of restitution, $e=0.6$, then find average impact force if it lasts for 0.2sec .

7. (a) Determine the acceleration of the body B and the tension in the string (Inextensible) supporting body A as shown. Pulleys are ideal pulley and Weights of body A and B are 500N and 750N respectively. Co-efficient of friction, $\mu = 0.2$. Inclination of plane with horizontal where the body B rests is $V:H=3:4$.

$V:H=3:4$.



(b) The 100kg wheel shown in fig. has a radius of gyration of 0.35m . A cable is wrapped around the central hub of the wheel and applying a horizontal force having variable magnitude of $P=(10+t)\text{N}$, where t is in seconds. If the wheel is initially at rest, determine its angular velocity in 5secs . Assume that the spool rolls without slipping.



Autumn (Odd) Semester Examination -2024

Course Code: PHYUGBS01

Course Title: Engineering Physics

Department of Appearing Students: Engineering, SEM-I, 1st Year

Full Marks: 80

Time: 3.00 hrs

Answer any eight questions $10 \times 8 = 80$

1. (a) Define i) unit vector and ii) null vector.
 (b) Find a unit vector parallel to the resultant of $\vec{r}_1 = 2\hat{i} - 4\hat{j} + 5\hat{k}$ and $\vec{r}_2 = \hat{i} + 2\hat{j} + 3\hat{k}$
 (c) Determine the value of "a" such that $\vec{A} = 2\hat{i} + a\hat{j} + \hat{k}$ and $\vec{B} = 4\hat{i} - 2\hat{j} - 2\hat{k}$ are mutually perpendicular to each other.
 (d) If $\vec{A} = 2\hat{i} - 3\hat{j} - \hat{k}$ and $\vec{B} = \hat{i} + 4\hat{j} - 2\hat{k}$ find $(\vec{A} + \vec{B}) \times (\vec{A} - \vec{B})$ (2+3+2+3)
2. (a) Define moment of inertia of an object. What is radius of gyration?
 (b) State the law of conservation of angular momentum.
 (c) Find the relation between torque and moment of inertia.
 (d) State parallel axes theorem of moment of inertia.
 (e) Three masses 3 kg, 4 kg and 5 kg are located at the corners of an equilateral triangle of side 1m. Find the center of mass of the system. ((1+1)+1+3+1+3)
3. (a) Define Young's modulus of elasticity. State Hooke's law of elasticity.
 (b) State Stoke's law. What is Reynold's number?
 (c) State Bernoulli's theorem.
 (d) Write down equation of continuity.
 (e) The length of a suspended wire increased by 10^{-4} of its original length when a stress of 10^7 Nm^{-2} is applied on it, Calculate the Young's modulus of the material of the wire. ((2+1)+(1+1)+2+1+2)
4. (a) Write down assumptions of the kinetic theory.
 (b) Deduce the perfect gas equation.
 (c) Find out the relation between rms speeds of the gas molecules with its temperature. (3+5+2)
5. (a) Write down Zeroth law of thermodynamics.
 (b) Write down second law of thermodynamics.
 (c) Find out the expression of the efficiency of a Carnot engine.
 (d) A Carnot engine takes in heat from a source at a temperature of 100°C and rejects heat to a sink at a temperature of 0°C . If the engine absorbs 1000 joules from the source, find the work done. (1+2+4+3)
6. (a) Define Interference of light.
 (b) Write down the conditions for observable interference pattern.
 (c) Calculate the fringe width using Young's double slit experiment.

OR

In Newton's ring experiment, the diameter of m-th dark ring is 8 mm and the diameter of (m+5)th dark ring is 12 mm. If the radius of curvature of the lower surface of the lens is 10m, find the wavelength of light used. (2+3+5)

7. (a) Find out the radius of a nucleus whose atomic mass number is 8.
 (b) What is Binding energy of a nucleus. Describe how Binding energy of a nucleon depends on the volume and the surface of a nuclei. (With proper mathematical description)
 (c) Show that the nuclear mass density is independent of Mass number.
 (d) State any two properties of nuclear force.

2+(1+2+2)+2+1

8. (a) Two protons with charge $1.6 \times 10^{-19} \text{ C}$ is separated within a nucleus at a distance $1.5 \times 10^{-15} \text{ m}$. Find the electrostatic repulsion force between the protons and the potential energy of the system as well
 (b) State the Gauss law of electrostatics and write its mathematical equation
 (c) Two particles with charge $+2e$ and $+3e$ separated at a distance 20 cm . Another particle with charge $1e$ is placed at a distance x from the particle with charge $+2e$ on the line of joining where no force is acting on it. Find the value of x

4+3+3

OR

- (a) What is a Bravais lattice? How many distinct types of Bravais lattices exist in 3-dimensional space?

- (b) Show that the Packing fraction of a Body-centered cubic (BCC) lattices is $\frac{\sqrt{3}\pi}{8}$.

- (c) A narrow beam of X-ray with wavelength $1.5 \text{ \AA}$ is reflected from an ionic crystal with an FCC lattice structure with a density of 3.32 gm/cm^3 , The molecular weight is 108 amu . Find the lattice constant?

2+3+5

9. (a) State the Heisenberg Uncertainty Principle.
 (b) The uncertainty of detecting the position of a particle is $\Delta x = 10 \text{ nm}$. Find the uncertainty of the momentum.
 (c) A metal whose work-function(ϕ) is 4.2 eV is irradiated by radiation whose wavelength is $200 \text{ \AA}$. Find the maximum Kinetic Energy of the emitted electrons?
 (d) Say, the matter wave of a quantum particle as: $\psi(x, t) = Ae^{i(kx - \omega t)}$. Say the energy operator $\hat{E} = i\hbar \frac{\partial}{\partial t}$ and momentum operator is $\hat{p} = -i\hbar \frac{\partial}{\partial x}$. Using the given operators show that, the energy of the particle is $E = \hbar\omega = h\nu$ and momentum of the particle $p = \hbar k = h/\lambda$

2+2+2+4

10. (a) What is photoelectric effect? Write down the Einstein's equation of Photoelectric effect. Draw the curve between the Stopping Potential (V_s) and the frequency of incident light in the Photo Electric Effect?
 (b) When Monochromatic light is shone on a clean metal surface, electrons are emitted from the surface due to the photoelectric effect. State and explain the effect on the emitted electrons of
 (i) increasing the frequency of the light.
 (ii) increasing the intensity of the light.
 (c) A Sodium Vapor lamp emits light photons with a wavelength of $5.89 \times 10^{-7} \text{ m}$. What is the energy of these photons?

4+4+2

11. (a) State de-Broglie hypothesis. Using the de-Broglie hypothesis, show that the wavelength λ of an alpha particle accelerated through a potential difference V is given by: $\lambda = \frac{h}{\sqrt{16m_p qV}}$, where m_p is the mass of a proton, q is the charge of a proton, and h is the Plank's constant.
- (b) What are isobars? How do isotopes of an element differ from one another?
- (c) The rate of decay of a radioactive substance is proportional to the number of nuclei present. Using $\frac{dN}{dt} = -\lambda N$, derive the formula $N = N_0 e^{-\lambda t}$, where N_0 is the initial number of nuclei and λ is the decay constant. Using the derived formula calculate the half-life ($t_{1/2}$)

3+2+5

Aliah University

Department of Electrical Engineering, B-Tech
Odd (Autumn) Semester Examination, 2024 (Reg./Supple)
Subject: Basic Electrical Engineering [EENUGES01]

Full Marks: 80
Year: I

Time: 3 Hours
Semester: II

- Instructions:**
1. Mention the question number clearly and write all the parts of a question at one place.
 2. Draw circuit/figure/phasor diagram wherever applicable (including numerical).
 3. Make suitable assumptions wherever necessary, symbols and notations have their usual meanings.

1. ☒ i) Distinguish between a Loop and Mesh of a circuit. 2x10= 20 Marks
☒ ii) State the principle of operation of a single-phase transformer.
☒ iii) What is series resonance?
☒ iv) Give two basic speed control schemes of DC shunt motor.
☒ v) State maximum power transfer theorem.
☒ vi) Is Ohm's law true for all conductors? Name two types of commercially available resistors.
☒ vii) An electric iron is rated 1000W, 240V. Find the current drawn and resistance of the heating element.
☒ viii) Can DC supply be applied to transformers?
☒ ix) What will happen if DC shunt motor is connected across AC supply?
☒ x) What are the three types of power used in AC circuits?

Answer any five questions

5x12=60 Marks

2. a) Write the steps involved in Norton's theorem and draw Thevenin's equivalent circuit across 20 Ω resistor for the shown in Fig. 1. 6
b) Differentiate between mesh and nodal analysis of a circuit. Determine the voltage through 10 Ω resistor using superposition theorem as shown in Fig. 2. 6

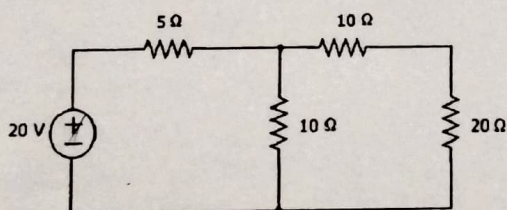


Fig. 1

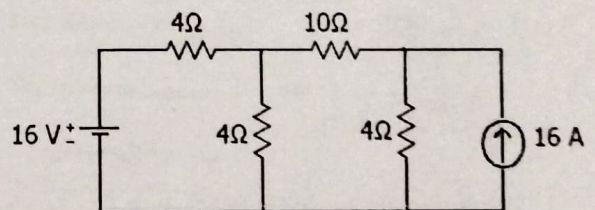


Fig. 2

3. a) In an R-C circuit with DC excitation (driven circuit) determine the expression for instantaneous capacitor voltage and instantaneous capacitor current and plot them with respect to time. 8
b) In an RL circuit having time constant 400 ms the inductor current decays and its value at 500 ms is 0.8 Amp. Find the equation of $i_L(t)$ for $t > 0^+$ 4
4. ☒ a) Derive the emf equation for single phase transformer. 6
b) A 50kVA, 5000/500V, 50 Hz, single phase transformer has the high and low voltage winding resistance of 8 ohm and 0.04 ohm, respectively. The no-load loss of the transformer is given as 900 W. Find the efficiency of the transformer at full load and 0.8 power factor lagging. 6

5. a) Given $v = 200 \sin 314t$ Volts and $i = 200 \sin (314t - 30^\circ)$ Amps for an AC circuit, determine
i) the power factor ii) True power iii) Apparent power iv) Reactive power. Indicate the unit of the power
calculated. **8**
- b) Write a short note on power triangle. **4**
6. a) With usual notations, develop the torque equation of a D.C. motor **6**
- b) A separately-excited DC generator develops an induced emf of 220 V while running at 900 rpm with a
flux of 0.05 Wb/pole. What would be required speed if the emf of 200 V is to be generated with a flux of
0.045 Wb? **6**

End-Semester Examination - 2024 (Autumn)
MATUGBS01
(Engineering Mathematics – I)
Semester – I (B. Tech in CEN, MEN, ECE, EEN, CSE/ BCA)
Full Marks – 80 **Time – 3 Hours**

Notations and symbols have their usual meaning.

Use separate answer sheets for Group – A and Group – B
Group–A (40 marks)

1. Answer all the following questions:

$1 \times 5 = 5$

- (a) Define convergent sequence in Real number.
- (b) Write the definition of oscillatory series.
- (c) Define local maxima and local minima of a function.
- (d) Write the statement of L'Hospital theorem.
- (e) State the relation between the Gamma and Beta function.

2. Answer all the following questions:

$1 \times 5 = 5$

- (a) The value of the $\lim_{x \rightarrow 1} \frac{1-x^2}{x^2+x-2}$ is
- | | |
|--------------------|---------|
| (1) -2 | (3) 1 |
| (2) $-\frac{2}{3}$ | (4) 2 |
- (b) The value of the $\beta(4, 7) =$
- | | |
|--------------------|---------------------|
| (1) $\frac{2}{99}$ | (3) $\frac{1}{33}$ |
| (2) $\frac{1}{99}$ | (4) $-\frac{4}{99}$ |
- (c) Find the value of c guaranteed by the Rolle's theorem of $f(x) = 3x^3 - 4x$ in $[-1, 0]$
- | | |
|-------------------|--------------------|
| (1) $\frac{2}{3}$ | (3) $-\frac{2}{3}$ |
| (2) $\frac{4}{9}$ | (4) $-\frac{4}{9}$ |
- (d) If a series $\sum x_n$ is convergent then which of the following always occurs
- | | |
|--|--|
| (1) $\lim_{n \rightarrow \infty} x_n = 1$ | (3) $\lim_{n \rightarrow \infty} x_n = 0$ |
| (2) $\lim_{n \rightarrow \infty} x_n = \infty$ | (4) $\lim_{n \rightarrow \infty} x_n \neq 0$ |

(e) Which of the following is not correct

(1) $\lim_{n \rightarrow \infty} \frac{1}{n}$ is convergent

(2) $\lim_{n \rightarrow \infty} \frac{1}{n}$ is divergent

(3) $\lim_{n \rightarrow \infty} \frac{1}{n^2}$ is convergent

(4) $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}}$ is divergent

3. Answer any **SIX** of the following questions:

$5 \times 6 = 30$

(a) Show that the below sequence is convergent

$4 + 1$

$\sqrt{3}, \sqrt{3 + \sqrt{3}}, \sqrt{3 + \sqrt{3 + \sqrt{3}}}, \sqrt{3 + \sqrt{3 + \sqrt{3 + \sqrt{3}}}}, \dots$. Find out where it converges.

(b) Test the convergence of the following series by Cauchy's root test.

5

$$\sum_{n=1}^{\infty} \left(\frac{n}{n+1} \right)^{n^2}$$

(c) Test the convergence of the series

5

$$\frac{2}{1 \cdot 3} + \frac{3}{3 \cdot 5} + \frac{4}{5 \cdot 7} + \dots$$

(d) Find the value of x for which the function $f(x) = x^3 - 6x^2 + 9x + 7$ has maximum and minimum. Find the global maximum of and minimum values in $[0, 6]$.

5

(e) (i) Show that $\lim_{x \rightarrow 0} \cos\left(\frac{1}{x}\right)$ does not exist.

$4 + 1$

(ii) Define the radius of convergence.

(f) Verify the Rolle's theorem of the following function

$f(x) = 2x^3 + x^2 - 4x - 2$, in the interval $-\sqrt{2} \leq x \leq \sqrt{2}$.

5

(g) Show that the function $f(x)$ is continuous at the points 6, 0, and -2, where,

$f(x) = |x| - |x + 2| + |x - 6|$.

5

(h) Evaluate by L' Hospital rule, $\lim_{x \rightarrow 1} \left(\frac{x}{x-1} - \frac{1}{\log x} \right)$.

5

(i) If m is a positive integers, then

5

$$\beta(m, n) = \frac{(m-1)!}{n(n+1)(n+2) \cdots (n+m-1)}$$

Group - B (40 marks)

1. Answer all the following questions:

$1 \times 5 = 5$

(a) The line $\frac{x-1}{2} = \frac{y-1}{1} = \frac{z-0}{-1}$ is perpendicular to the line

(1) $\frac{x}{1} = \frac{y}{-1} = \frac{z}{1}$

(2) $\frac{x-2}{1} = \frac{y-5}{1} = \frac{z-0}{1}$

(3) $\frac{x-1}{2} = \frac{y-1}{-3} = \frac{z-7}{1}$

(4) $\frac{x-2}{0} = \frac{y-4}{1} = \frac{z-0}{-1}$

(b) The distance between two planes $12x - 4y + 3z + 10 = 0$ and $12x - 4y + 3z + 36 = 0$ is

(1) 1

(2) 2

(3) 3

(4) 4

(c) The angle between two planes $x + y + 1 = 0$ and $y + z + 1 = 0$ is

(1) 0

(2) $\frac{\pi}{2}$

(3) $\frac{\pi}{3}$

(4) $\frac{\pi}{4}$

(d) If $f(x, y, z) = 3x^2y - y^3z^2$, then $\text{grad } f$ at $(1, -2, -1)$ is equal to

(1) $12i + 9j + 16k$

(2) $-12i - 9j - 16k$

(3) $i + 9j + 16k$

(4) $i + 9j - 16k$

(e) For which type of curve can we use Green's theorem?

(1) open curve

(2) closed curve

(3) both for open and closed

(4) all of above

2. Answer all the following questions:

$1 \times 5 = 5$

(a) Convert the point $(7, -7\sqrt{3})$ in to the polar coordinate.

(b) Find out the direction cosine of the straight line $\frac{2x-1}{6} = \frac{2-y}{4} = \frac{2z-3}{24}$.

(c) What do you mean by the curvature of a curve?

(d) What do you mean by the arc length of a curve?

(e) Find out the value of the double integral $\int_0^1 \int_0^1 (x^2) dx dy$

3. Answer any **SIX** of the following questions:

$5 \times 6 = 30$

(a) I) If a line makes an angle $\frac{\pi}{3}$ with x -axis, $\frac{\pi}{3}$ with y -axis and $\theta \in (0, \pi)$ with z -axis, then find out the value of θ .

$2\frac{1}{2}$

II) Find out the nature of the quadratic surface $2x^2 + 5y^2 + 3z^2 - 4x + 20y - 6z = 5$ and hence evaluate its volume.

$2\frac{1}{2}$

(b) Find out the image of the point $(-3, 8, 4)$ with respect to the plane $6x - 3y - 2z + 1 = 0$.

5

(c) Write down the equation of the line that is perpendicular to both lines given below; hence, find out the distance between the lines given below

5

$$\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-0}{0} \text{ and } \frac{x-1}{0} = \frac{y-1}{1} = \frac{z-2}{1}$$

- (d) Find out the equation of the plane passing through the point $(1, 1, 2)$ and perpendicular to the planes $x - y + z = 1$ and $2x - 3y + z = 1$. 5

- (e) Find out the curvature at any point t of the curve 5

$$F(t) = (\cos t, -\sin t, 2t).$$

- (f) Find out the torsion of the curve 5

$$F(t) = (e^t, e^{-t}, \sqrt{2}t).$$

- (g) Evaluate

$$\iint (x + y) dx dy$$

over the region enclosed by the circle of radius 2, $y = x$ and the positive x axis. 5

- (h) Evaluate

$$\iiint (x + y + z + 1) dx dy dz$$

over the region defined by $x \geq 0$, $y \geq 0$, $z \geq 0$, $x + y + z \leq 1$. 5

- (i) Use Green's theorem to evaluate

$$\oint_C (3xy^2 - 6y) dx + (y - 3xy) dy$$

where C is the boundary of the region $x = 0$, $y = 0$, $2x + 3y = 12$. 5

- (j) Use Divergence theorem to evaluate

$$\iiint_S \vec{F} \cdot d\vec{S}$$

where $\vec{F} = \cos(\pi x)\hat{i} + 2xy^3\hat{j} + (z^2 - 4x)\hat{k}$ and S is the surface of the box with $-1 \leq x \leq 2$, $0 \leq y \leq 1$ and $1 \leq z \leq 4$. 5

Full Marks: 80

Time: 3:00Hrs.

ARABIC

Q. 1. Attempt any ten questions, each question carries equal marks. 2x10=20

- i. Identify the letters of the following words and split them: يَسْتَكْبِرُ، يُسْتَعْتَبُونَ
- ii. Read the following words and write the pronunciations in English: يُمْسِكُهُنَّ، فَلْنُحْيِيَنَّهٗ
- iii. Merge the following letters to make word. م+س+ت+ق+ي+م، ي+س+ت+ط+ي+ع+و+ن
- iv. Identify the following words either definite (المعرفة) or indefinite (النكرة) : بَيْتٌ، الْبُرُوجُ، يَا كَرِيمُ، جَسْرٌ
- v. Identify the following words as they are 'Noun' (اسم) or 'Verb' (فعل) : بِسْمِ اللَّهِ، لَنْ نَبْرَحَ، أَلْحَمْدُ، قَدْ نَرَى
- vi. How many kinds of sentences are there in Arabic?
- vii. Change the Gender of following words: طَبِيبٌ، صَفْرَاءٌ، مُهَنْدِسٌ، مُؤَمِّنَةٌ
- viii. Write any four demonstrative Pronouns (اسم الإشارة) of Arabic.
- ix. How many kinds of Madd are in Arabic? And what are they?
- x. What do you Know about Tanween? write in short with examples.
- xi. Change the Nominal sentence into verbal sentence and Verbal sentence into Nominal Sentence as well. قَرَأَ الطَّالِبُ الْكِتَابَ، أَلْبِنْتُ رَجَعْتُ مِنَ الْمَدْرَسَةِ

Q. 2. Attempt any two questions, each of them carries equal marks. 10x2=20

- i. Write about Gender in Arabic in details by citing of examples.
- ii. Write the Arabic numbers 1-20 in Arabic scripts.
- iii. Write down the Moon Letters (الحروف القمرية) and Sun Letters (الحروف الشمسية) in Arabic.

Islamic Studies

Answer any four Questions:

10x4=40

1. Write a short note about the life of Prophet Muhammad (PBUH).
2. How Many Pillars are there in "Islam"? Write in brief.
3. What is Tawheed? Define its types with example.
4. What do You Know about Zakat? To whom Zakat is Farz? Write a short note on it.
5. How Many Pillars of Iman? Write about all of them in short.